

SELF-SUPERVISED PROTOTYPE-BASED PREDICTION OF FUTURE CRYPTOCURRENCY PRICE STRUCTURES USING K-LINE CHART EMBEDDINGS

Technical Project Concept Paper

A proposed framework for predicting the approximate visual and structural dynamics of the next 30-candle market window

Student	[Student Name]
Student ID	[Student ID]
Programme	Master of [Programme Name]
Department	[Department Name]
Institution	[University Name]
Supervisor	[Supervisor Name]
Date	[Month and Year]

Abstract

Cryptocurrency forecasting remains difficult because market behaviour is volatile, non-stationary and influenced by changing liquidity, sentiment and external conditions. Most forecasting systems attempt to predict an exact price, a return, or a manually defined direction such as upward or downward movement. These targets may not adequately represent the internal structure of an extended future price window. This concept paper proposes a prototype-based prediction framework that forecasts the approximate visual and structural dynamics of the next cryptocurrency market window. Historical opening, high, low and close (OHLC) observations will be divided into fixed windows containing 30 consecutive candles and rendered as standardised K-line chart images. A contrastive self-supervised convolutional neural network will learn embeddings from the images without manually assigned market-pattern labels. The training embeddings will then be grouped into an empirically selected number of clusters. Each cluster will represent a recurring category of 30-candle price behaviour, and its medoid will serve as an observable historical prototype. For supervised learning, the cluster assigned to the immediately following 30-candle window will become the prediction target of the current window. The resulting model will estimate a probability distribution over the discovered clusters and return the medoid of the most probable cluster as an approximate reference for the next window. The framework therefore combines self-supervised visual representation learning, unsupervised market-structure discovery, medoid-based interpretation and supervised transition prediction. It will be evaluated through chronological out-of-sample testing, clustering measures, classification measures, structural-similarity measures and optional paper-trading analysis.

Keywords: cryptocurrency forecasting; K-line charts; candlestick charts; self-supervised learning; contrastive learning; convolutional neural networks; clustering; medoids; prototype-based prediction; market structures.

Table of Contents

Abstract.....	2
Project at a Glance.....	4
1. Introduction.....	5
2. Aim, Objectives and Research Questions.....	7
3. Scope, Assumptions and Boundaries.....	9
4. Proposed Conceptual Framework.....	10
5. Detailed Methodology.....	12
6. Training, Validation and Leakage Control.....	17
7. Evaluation Plan.....	18
8. Trading-Oriented Interpretation.....	20
9. Expected Novelty and Contribution.....	21
10. Proposed Software and System Design.....	22
11. Risks, Challenges and Mitigation.....	23
12. Ethical, Reproducibility and Practical Considerations.....	24
13. Deliverables and Work Plan.....	25
14. Success Criteria.....	26
15. Anticipated Limitations.....	27
16. Future Extensions.....	28
17. Conclusion.....	29
References.....	30

Project at a Glance

Item	Current Project Definition
Project type	Technical research and machine-learning development project
Primary objective	Develop a trading-oriented prediction model that estimates the approximate price structure and visual dynamics of the next market window.
Target market	One cryptocurrency: [to be selected]
Raw input	Historical opening, high, low and close values at a fixed interval: [to be selected]
Frame definition	One K-line image containing 30 consecutive candles
Representation method	Contrastive or related self-supervised CNN embedding model
Unsupervised stage	Cluster the learned chart embeddings into K latent structural groups
Prototype	The medoid: an actual historical chart nearest to the centre of each cluster
Prediction target	Cluster label of the immediately following non-overlapping 30-candle window
User-facing output	Cluster probabilities, predicted cluster and its representative medoid
Interpretation	Approximate structural reference, not an exact future OHLC path or guaranteed trade signal

Table 1. Summary of the proposed project.

The project reframes cryptocurrency prediction as a future-window prototype classification problem. Rather than forcing the next market period into a manually specified bullish or bearish label, the framework discovers recurring 30-candle structures from the data and then learns to predict which discovered structure is most likely to occur next.

1. Introduction

1.1 Background

A fundamental challenge in modelling complex systems is explaining how observable local structures can coexist with aggregate behaviour that remains difficult to predict. Nonlinear systems may be sensitive to initial conditions, interactions and incomplete knowledge. Chaos theory provides a useful conceptual background for understanding why deterministic components do not automatically produce reliable long-term forecasts (Oestreicher, 2007). Financial markets, however, should not be treated as purely deterministic chaotic systems because their behaviour also reflects random events, information arrival, changing expectations and external economic conditions.

Financial prediction is therefore primarily probabilistic. A model identifies relationships within historical observations and estimates probable future outcomes rather than guaranteeing a single trajectory. Traditional financial forecasting has relied on econometric models, technical indicators, statistical analysis and expert judgement. These approaches remain valuable, but their assumptions may limit their ability to represent nonlinear interactions and changing market regimes.

Machine-learning and deep-learning techniques have expanded the available modelling approaches. Recent cryptocurrency studies have applied recurrent neural networks, regression methods, ensemble learning and hybrid optimisation systems to historical numerical data. Rodrigues and Machado (2025), for example, compared several machine-learning approaches for high-frequency cryptocurrency forecasting, while Zhu et al. (2023) proposed a hybrid Bitcoin-price framework combining feature selection, optimisation and support-vector regression.

Cryptocurrency markets intensify the forecasting problem because they trade continuously and can experience rapid changes in volatility, liquidity, sentiment and regulatory expectations. A relationship learned during one period may weaken when the market enters another condition. Consequently, a model may perform well during one historical regime and poorly in a later one.

Many forecasting studies define the target as an exact future price, future return or direction. These targets are convenient but may oversimplify the behaviour inside a future period. Two 30-candle windows can finish with the same return while exhibiting very different paths. One may show a steady trend, while the other may contain a sharp decline followed by a recovery. A single directional label would treat both windows as equivalent.

The importance of examining data structure is illustrated by Anscombe's quartet. Anscombe (1973) showed that datasets with nearly identical statistical summaries can display substantially different visual patterns. The example does not establish that image-based financial models will necessarily outperform numerical models. It instead demonstrates that the arrangement of observations may communicate structure that selected summaries do not express fully.

Candlestick or K-line charts provide a compact visual representation of OHLC data. Each candle encodes the opening, high, low and closing prices for one interval. A sequence of candles displays relative direction, range, volatility and temporal arrangement. Previous studies have demonstrated that financial time series can be encoded or rendered as images and processed by convolutional neural networks (Chen & Tsai, 2020; Hung & Chen, 2021; Kusuma et al., 2019).

A K-line image does not introduce new raw information because it is derived from OHLC values. Its potential benefit lies in representation. A CNN may learn local spatial relationships among candle bodies, wicks, relative positions and sequences. The research question is therefore whether the image representation and the inductive bias of the CNN produce useful structural features, not whether the image contains information unavailable in the numerical data.

Dimension	Conventional formulation	Proposed formulation
Target	Exact future price, return, or manual direction label	Data-driven cluster of the next 30-candle window
Label source	Specified by the researcher before training	Discovered from self-supervised embeddings
Forecast horizon	Often one next value or candle	Complete next non-overlapping 30-candle structure
Output	Number or abstract class	Probability distribution plus an observed medoid chart
Interpretation	Point or directional forecast	Approximate prototype of future price dynamics

Table 2. Conceptual distinction between common forecasting formulations and the proposed approach.

1.2 Problem Statement

Existing cryptocurrency-prediction models frequently formulate forecasting as regression or simple classification. Regression models may imply unrealistic precision by estimating an exact future value. Directional classifiers reduce that precision but discard much of the internal structure of the future period.

Image-based financial models have shown that CNNs can process candlestick representations, but many existing approaches depend on manually defined pattern labels or direct price-direction targets. Such formulations restrict the model to categories selected before the data's latent structures have been examined.

There is therefore a need for a framework that learns chart representations without manual pattern labels, discovers recurring market-window structures from historical data, represents each structure through an actual example and predicts the probable structural category of the next window. The project addresses this need by combining contrastive self-supervised learning, clustering, medoid selection and supervised next-window prediction.

2. Aim, Objectives and Research Questions

2.1 Main Aim

The main aim is to develop and evaluate a trading-oriented prediction model that uses the current 30-candle K-line window to estimate an approximate representation of the price structure and visual dynamics of the immediately following 30-candle window.

The model will not attempt to reproduce the exact OHLC value of every future candle. It will predict the cluster to which the next window is most likely to belong and return the medoid of that cluster as a structural visual reference.

2.2 Specific Objectives

1. Collect, validate and preprocess historical OHLC data for one selected cryptocurrency.
2. Divide the chronological time series into fixed windows containing 30 consecutive candles.
3. Develop a consistent procedure for converting every window into a standardised K-line chart image.
4. Train a contrastive self-supervised CNN to learn chart embeddings without manually assigned pattern labels.
5. Cluster the training embeddings into an empirically selected number of recurring market-structure groups.
6. Identify the medoid of each cluster as its representative historical price-structure prototype.
7. Construct supervised pairs in which the current window predicts the cluster of the immediately following window.
8. Train a classifier to estimate the probability that the next window belongs to each discovered cluster.
9. Return the predicted cluster medoid as an approximate visual reference for the expected next-window dynamics.
10. Evaluate the complete framework using chronological out-of-sample tests, baselines and ablation studies.
11. Investigate whether the resulting predictions can support a carefully constrained paper-trading or decision-support experiment.

2.3 Research Questions

1. Can contrastive self-supervised learning produce meaningful embeddings from 30-candle cryptocurrency K-line images?
2. Can the embeddings be divided into stable and interpretable clusters representing recurring price-window structures?
3. What number of clusters provides an appropriate balance among separation, stability, sample size and interpretability?
4. Can the next window's cluster be predicted from the current window more accurately than simple baseline methods?
5. Does the medoid of the predicted cluster provide a useful approximation of the actual next window's structure?

6. Does self-supervised pretraining improve prediction compared with direct supervised learning or random features?
7. Which discovered structures are most predictable, and which are associated with the greatest uncertainty?

Research question	Related implementation objective	Primary evidence
RQ1	Train a contrastive self-supervised CNN to learn chart embeddings without manually assigned pattern labels.	Embedding quality and retrieval
RQ2	Cluster the training embeddings into an empirically selected number of recurring market-structure groups.	Cluster validity and stability
RQ3	Identify the medoid of each cluster as its representative historical price-structure prototype.	Model selection across K
RQ4	Construct supervised pairs in which the current window predicts the cluster of the immediately following window.	Classification against baselines
RQ5	Train a classifier to estimate the probability that the next window belongs to each discovered cluster.	Structural-distance evaluation
RQ6	Return the predicted cluster medoid as an approximate visual reference for the expected next-window dynamics.	Ablation study
RQ7	Evaluate the complete framework using chronological out-of-sample tests, baselines and ablation studies.	Per-cluster error analysis

Table 3. Alignment of research questions with implementation and evidence.

3. Scope, Assumptions and Boundaries

3.1 Current Scope

- One cryptocurrency: [to be selected].
- One fixed candle interval: [to be selected, for example 5-minute, 15-minute, 1-hour or daily candles].
- OHLC variables as the principal raw input.
- Thirty candles per image frame.
- One current frame used to predict the immediately following non-overlapping frame.
- Self-supervised CNN embeddings, unsupervised clustering and supervised next-cluster classification.
- A research prototype and paper-trading evaluation rather than direct real-money execution.

3.2 Key Assumptions

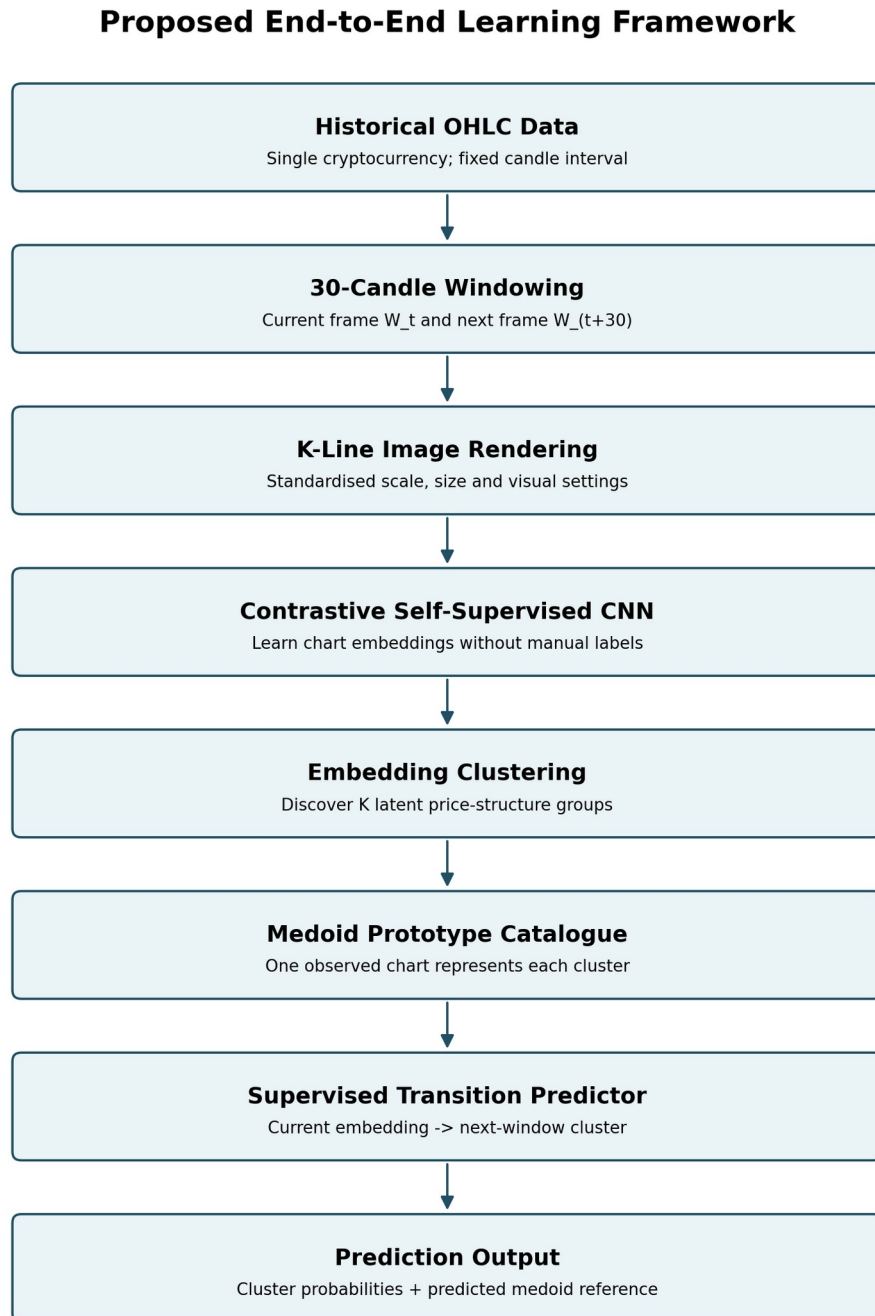
- Historical price structures contain recurring similarities that can be represented in a learned embedding space.
- Some transitions between discovered structures contain limited but measurable predictability.
- Normalisation can reduce unwanted dependence on absolute historical price levels while preserving relevant shape information.
- A cluster medoid can provide a more interpretable representative than a synthetic centroid.
- Chronological evaluation can provide a defensible estimate of out-of-sample performance, although market non-stationarity will remain.

3.3 Exclusions from the Initial Version

- Guaranteed price or profit prediction.
- Automatic live order execution.
- Multi-asset generalisation claims.
- Order-book, news, sentiment and macroeconomic variables in the primary model.
- Manual assignment of named candlestick patterns to the clusters.
- Reclustering the test data independently from the training data.

4. Proposed Conceptual Framework

The proposed system contains three connected learning stages: self-supervised representation learning, unsupervised market-structure discovery and supervised next-window prediction. The learned CNN embedding provides the link among the stages.



The model predicts an approximate structural prototype, not an exact future price path.

Figure 1. Proposed end-to-end project architecture.

4.1 Stage One: Self-Supervised Representation Learning

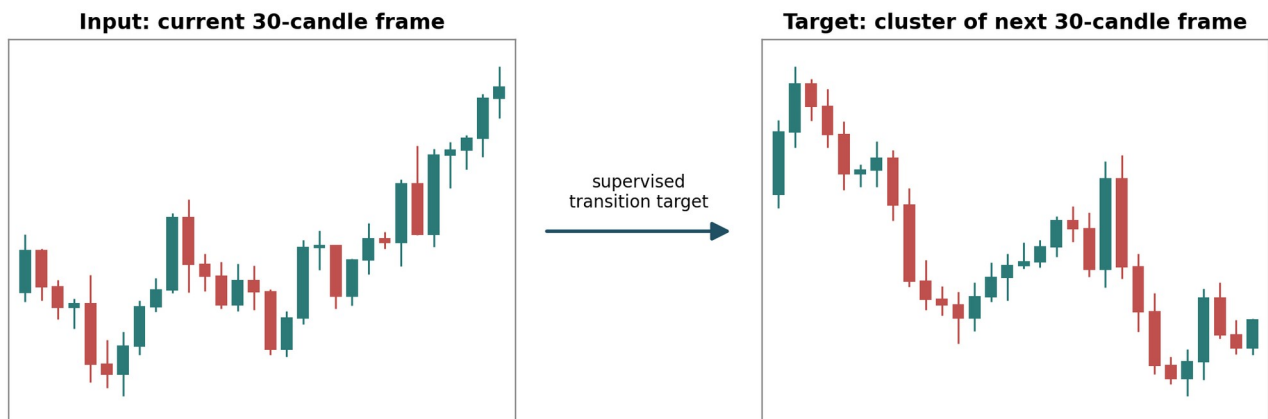
A contrastive CNN will learn a feature vector for each K-line image without requiring a manually assigned bullish, bearish or named-pattern label. Two meaning-preserving views of the same image will form a positive pair. The encoder will be trained to bring positive-pair representations closer while separating representations of other windows. SimCLR provides a foundational contrastive-learning framework and shows that augmentation design is central to the representations that are learned (Chen et al., 2020).

4.2 Stage Two: Unsupervised Structure Discovery

After self-supervised training, the encoder will transform every training chart into a fixed-dimensional embedding. An unsupervised clustering algorithm will divide the embeddings into K groups. DeepCluster demonstrates the broader feasibility of combining learned visual features with clustering to discover latent structures in unlabelled images (Caron et al., 2018). In this project, the groups are expected to correspond to recurring forms of price-window behaviour rather than conventional object classes.

4.3 Stage Three: Supervised Next-Window Prediction

Every 30-candle window will receive a cluster label. The label of the immediately following window becomes the supervised target for the current window. The classifier therefore learns a transition from the current structure to the probable structural group of the next period.



Illustrative synthetic chart windows; values are not market observations.

Figure 2. Construction of a current-window and next-window supervised pair.

5. Detailed Methodology

5.1 Data Acquisition

Historical OHLC observations will be obtained from a reliable exchange or market-data source. The selected source, asset, interval and study period must be documented exactly so that the experiment can be reproduced.

Parameter	Planned value	Reason it must be fixed
Cryptocurrency	[To be selected]	Defines the market population studied.
Exchange or provider	[To be selected]	Different venues may have different prices, liquidity and missing intervals.
Candle interval	[To be selected]	Determines the real-time horizon represented by 30 candles.
Study period	[Start date - end date]	Defines the historical regimes available for training and testing.
Timezone	UTC or provider standard	Prevents inconsistent candle boundaries.
Window length	30 candles	Core design decision supplied for the project.
Window stride	30 candles initially	Creates non-overlapping frames and reduces dependence between adjacent samples.

Table 4. Data parameters that must be finalised before implementation.

Each record will be checked for duplicate timestamps, missing intervals and invalid price relationships. A valid candle must satisfy $H_t \geq \max(O_t, C_t)$ and $L_t \leq \min(O_t, C_t)$. Missing candles should not be silently interpolated unless a documented experiment justifies that choice.

5.2 Window Construction

Let each candle at time t be represented by the tuple:

$$x_t = (O_t, H_t, L_t, C_t)$$

For a fixed window length $L = 30$, the current frame beginning at t is:

$$W_t = \{x_t, x_{t+1}, \dots, x_{t+L-1}\}$$

The immediately following non-overlapping frame is:

$$W_{t^+} = \{x_{t+L}, x_{t+L+1}, \dots, x_{t+2L-1}\}$$

Using a stride of 30, the first pair is candles 1-30 predicting the cluster of candles 31-60. The next pair is candles 31-60 predicting candles 61-90. Each complete supervised example therefore spans 60 candles.

Sample	Input frame	Target frame	Supervised target
1	Candles 1-30	Candles 31-60	Cluster assigned to candles 31-60
2	Candles 31-60	Candles 61-90	Cluster assigned to candles 61-90
3	Candles 61-90	Candles 91-120	Cluster assigned to candles 91-120

Table 5. Example construction of non-overlapping current and future windows.

5.3 Price Normalisation

Normalisation is necessary because the desired clusters should primarily reflect price structure rather than the absolute historical price level. Without normalisation, windows from early low-price periods could be separated from later high-price periods even where their relative shapes are similar.

A suitable primary transformation is to express prices relative to the first opening price of each window:

$$P_tilde(t,j) = \log(P_ (t,j) / O_ (t,1))$$

This transformation removes the absolute level while preserving proportional movement, direction and range. A min-max alternative may be tested, but it can remove information about the magnitude of volatility. The normalisation method should therefore be treated as an experimental choice.

5.4 K-Line Chart Rendering

Every window will be rendered with identical visual settings. Axes, timestamps, price labels, watermarks and legends should normally be removed so that the model does not learn irrelevant text or chart artefacts. The chart dimensions, candle spacing, wick width, body width, background and direction encoding must remain consistent.

Rendering element	Recommended rule	Reason
Image size	Fixed, for example 128 x 128 or 224 x 224 pixels	Ensures consistent CNN input.
Number of candles	Exactly 30	Preserves the project horizon.
Axes and labels	Remove	Prevents learning dates and absolute price labels.
Vertical scale	Derived from normalised window values	Makes shape comparable across periods.
Colours	Fixed or grayscale experiment	Tests whether geometry rather than arbitrary colour drives learning.
Whitespace	Standardised and minimised	Reduces irrelevant pixels.
File format	Lossless PNG	Avoids compression artefacts.

Table 6. Proposed K-line rendering controls.

5.5 Contrastive Self-Supervised CNN

A compact CNN such as ResNet-18 is a suitable starting encoder. A projection head will be used during contrastive training and removed when embeddings are generated for clustering and supervised prediction. The encoder output before the projection head will serve as the chart representation.

For each original image I_i , two financially valid transformations A_1 and A_2 will create the positive views $v_i^{(1)}$ and $v_i^{(2)}$. The transformations must preserve the time order, price direction and underlying OHLC meaning.

Augmentation	Initial decision	Rationale
Small changes in line thickness	Candidate	Changes rendering style without changing price structure.
Mild pixel noise or blur	Candidate	Encourages robustness to minor rendering differences.
Minor brightness variation	Candidate with caution	Acceptable only when brightness does not encode market direction.
Limited masking	Candidate with caution	Must not remove complete temporal sections or critical candles.
Horizontal flip	Prohibited	Reverses the direction of time.
Vertical flip	Prohibited	Reverses price direction.
Arbitrary rotation	Prohibited	Changes the meanings of the axes.
Aggressive crop	Prohibited	Can remove candles or alter the window horizon.
Candle reordering	Prohibited	Destroys temporal structure.

Table 7. Candidate and prohibited contrastive augmentations.

A normalised temperature-scaled cross-entropy objective can be used. In conceptual form, the loss increases similarity between the embeddings of the two views of the same chart and decreases similarity to other charts in the batch. The temperature, embedding dimension, batch size and augmentation composition will be tuned using only the training and validation periods.

5.6 Embedding Clustering

After self-supervised training, the frozen encoder will generate an embedding h_t for each training image. The primary clustering method will be K-medoids because it directly identifies an observed member as the representative of each cluster. Alternative algorithms may be included as comparisons.

$$h_t = f_{\theta}(I_t)$$

$$C = \{C_1, C_2, \dots, C_K\}$$

For cluster C_k , its medoid is the member that minimises the total distance to all other members:

$$m_k = \arg \min_{(i \in C_k)} \sum_{(j \in C_k)} d(h_i, h_j)$$

Because the medoid is an actual chart rather than a synthetic mean, it can be displayed directly and linked to historical descriptive statistics. Kaufman and Rousseeuw (1990) provide the foundational treatment of medoid-based clustering.

Candidate measure	Desired interpretation	Use in selection
Silhouette score	Cohesion within clusters versus separation from neighbouring clusters	Higher values generally indicate clearer grouping.
Davies-Bouldin index	Average similarity between each cluster and its most similar alternative	Lower values are preferred.

Candidate measure	Desired interpretation	Use in selection
Calinski-Harabasz score	Between-cluster dispersion relative to within-cluster dispersion	Higher values are preferred.
Cluster-size distribution	Whether groups contain sufficient examples	Avoid highly fragmented solutions.
Stability across seeds	Whether similar solutions recur	Reject clusters that are highly unstable.
Medoid interpretability	Whether members resemble their prototype	Qualitative diagnostic, not the sole criterion.
Downstream predictability	Whether transitions can be learned	Secondary criterion after unsupervised validity.

Table 8. Criteria for selecting the number and quality of clusters.

The number of clusters K will be selected from a candidate range. A very small K may combine distinct structures, whereas a very large K may create small classes that cannot be predicted reliably. Rousseeuw's (1987) silhouette method will provide one quantitative diagnostic, but the final choice should combine separation, stability, size and interpretability.

5.7 Cluster and Medoid Catalogue

A catalogue will be created for each discovered cluster. Neutral identifiers such as Cluster 1 and Cluster 2 will be used during modelling. Descriptive names may be assigned after examining the cluster, but the names must not influence the unsupervised grouping.

Catalogue field	Purpose
Cluster identifier	Stable reference used by the classifier.
Medoid image and embedding	Observable prototype and machine-readable representation.
Number and percentage of members	Describes prevalence and class imbalance.
Median starting-to-ending return	Summarises broad direction without defining the cluster.
Median volatility and total range	Describes movement magnitude.
Trend slope and direction changes	Supports post-hoc structural interpretation.
Within-cluster distance distribution	Indicates the diversity around the medoid.
Historical transition probabilities	Provides a simple Markov baseline and interpretive context.

Table 9. Proposed fields in the medoid prototype catalogue.

5.8 Supervised Next-Window Classifier

The target of the current window is the cluster label of the following window. If c_t represents the cluster assigned to W_t , then the supervised target is:

$$y_t = c_{(t+L)}, \text{ where } L = 30 \text{ candles}$$

The classifier g_{ψ} will estimate a probability distribution over the K clusters:

$$P(y_t = k \mid h_t), \text{ for } k = 1, 2, \dots, K$$

A multilayer perceptron is an appropriate primary classifier. Logistic regression, support-vector machines and gradient-boosted trees may be used as comparisons. An extended temporal experiment may use several previous window embeddings and a GRU, LSTM or Transformer, but the core experiment should first evaluate one-window-to-next-window predictability.

5.9 Prediction Output and Prototype Rescaling

The system will return the full cluster-probability vector rather than only a label. The most probable cluster will identify the primary medoid, while the second-most probable cluster may be displayed when uncertainty is substantial.

An optional display can rescale a medoid's relative path so that it begins near the latest observed closing price. If $r_{(m,j)}^P$ is the medoid's relative log-price sequence and C_t is the current closing price, an illustrative rescaled prototype is:

$$\hat{P}_{(t+j)}^P = C_t * \exp(r_{(m,j)}^P)$$

The rescaled prototype must be labelled as an approximate structural reference. It is not a point forecast and is not expected to reproduce every future candle.

6. Training, Validation and Leakage Control

6.1 Chronological Division

The raw time series must be divided chronologically before any learned component is fitted. A possible initial split is 70% training, 15% validation and 15% test, although the final proportions should reflect the amount of available data.

Randomly distributing windows across the splits would be inappropriate because adjacent periods are dependent and overlapping samples may share underlying candles. A chronological design better reflects deployment, where the model is trained on the past and evaluated on a later unseen period.

6.2 Training-Only Operations

The following components must be fitted using only the training period:

- any global normalisation parameters;
- self-supervised CNN weights;
- embedding-dimension or augmentation selection;
- the clustering solution and selected K;
- cluster medoids;
- the supervised classifier and class weights;
- probability-calibration parameters.

Validation and test images will be encoded using the training encoder and assigned to the nearest training medoid. The test observations must not be independently reclustered, because doing so would change the definition of the prediction classes.

6.3 Boundary Purging and Overlap Control

Any sample whose current window and target window fall into different dataset partitions must be removed. With 30-candle inputs and 30-candle targets, each complete pair covers 60 candles. If rolling windows with a smaller stride are later tested, a purge gap must ensure that candles represented in a training sample do not reappear within validation or test samples.

6.4 Walk-Forward Evaluation

A walk-forward study is recommended in addition to one final split. The system can be trained on an initial historical period, tested on the next period, moved forward and refitted. This will reveal whether performance is concentrated in one regime or remains reasonably consistent across several market conditions.

7. Evaluation Plan

The framework should be evaluated at four levels: representation quality, cluster validity, next-cluster classification and structural approximation. Optional paper-trading analysis should be reported separately from the main machine-learning evaluation.

Evaluation level	Core measures	Question answered
Representation	Nearest-neighbour retrieval, linear probe, embedding visualisation, augmented-view similarity	Do the embeddings organise similar chart structures meaningfully?
Clustering	Silhouette, Davies-Bouldin, Calinski-Harabasz, cluster stability and size	Are the discovered market structures coherent and reproducible?
Classification	Balanced accuracy, macro-F1, per-class recall, log loss, Brier score and confusion matrix	Can the next structure be predicted better than baselines?
Structural approximation	Embedding distance, path DTW, return, slope, volatility and range differences	Is the predicted medoid close to the actual next-window dynamics?
Paper trading	Net return, drawdown, turnover, hit rate and transaction-cost sensitivity	Could the structural output support a cautious decision rule?

Table 10. Multi-level evaluation framework.

7.1 Baseline Models

Baseline	Definition	Purpose
Majority cluster	Always predict the most frequent training target.	Controls for class imbalance.
Persistence	Predict that the next window remains in the current cluster.	Tests whether the model learns more than regime persistence.
First-order Markov model	Predict from the training transition matrix $P(\text{next cluster} \text{current cluster})$.	Provides a simple state-transition benchmark.
Numerical OHLC model	Train a classifier on equivalent normalised OHLC values.	Tests whether image representation adds modelling value.
Direct supervised CNN	Train the image encoder only with next-cluster labels.	Measures the contribution of self-supervised pretraining.
Random encoder	Use untrained CNN features.	Tests whether performance comes from learned representations.

Table 11. Proposed baseline models and their diagnostic purpose.

7.2 Ablation Studies

- Contrastive pretraining versus direct supervised training.
- K-medoids versus k-means or hierarchical clustering.
- Cosine distance versus Euclidean distance.
- Log-relative normalisation versus min-max normalisation.
- Colour charts versus grayscale charts.
- Different embedding dimensions and encoder capacities.
- Window lengths of 15, 30 and 60 candles.
- Different values of K.
- Different valid augmentation combinations.

- Frozen encoder versus supervised fine-tuning.
- Single preceding window versus several preceding windows.
- Non-overlapping versus rolling-window sampling.

8. Trading-Oriented Interpretation

The project objective is to develop a trading prediction model, but the direct machine-learning target is a structural prototype rather than a buy or sell instruction. A separate interpretation layer may associate each cluster with historical statistics such as median return, probability of positive return, volatility and maximum drawdown.

A paper-trading decision rule could act only when the predicted cluster has a sufficiently strong historical directional tendency and the classifier's probability exceeds a threshold selected on validation data. Low-confidence predictions should produce a no-trade decision. Any simulation must include transaction fees, slippage and turnover.

Possible output condition	Research-only interpretation
High probability for a historically positive-trend cluster	Potential long signal, subject to confidence, risk and cost rules.
High probability for a historically negative-trend cluster	Potential defensive or short signal only where the study design permits.
High uncertainty or mixed prototype probabilities	No-trade or reduced-exposure outcome.
Large distance from all training medoids	Out-of-distribution warning; avoid forcing a confident forecast.

Table 12. Example separation between structural prediction and trading interpretation.

The project's academic claims should focus first on representation, clustering and predictive performance. A profitable paper-trading result would be supplementary evidence, not a substitute for rigorous out-of-sample evaluation.

9. Expected Novelty and Contribution

The novelty should not be described simply as using CNNs or candlestick images, because these techniques have appeared in earlier financial research. The expected contribution lies in the particular integration of the following components:

1. Self-supervised visual representation learning for fixed 30-candle cryptocurrency windows.
2. Data-driven discovery of market-window categories rather than manual bullish, bearish or named-pattern labels.
3. Use of a medoid as an observable historical prototype for each latent group.
4. Supervised prediction of the cluster assigned to the immediately following complete window.
5. A model output consisting of cluster probabilities and an approximate future structural prototype.
6. Evaluation that combines clustering, classification and structural-similarity evidence.

A defensible contribution statement is:

This study proposes a prototype-based cryptocurrency forecasting framework in which contrastively learned K-line embeddings are used to discover latent market-window structures, after which supervised learning predicts the medoid-referenced cluster of the immediately following window.

Any claim of novelty should be qualified with wording such as 'to the best of the researcher's knowledge, based on the reviewed literature' until the final systematic literature review has been completed.

10. Proposed Software and System Design

The implementation should separate data preparation, model training, clustering, supervised prediction and evaluation into reproducible modules. Python is suitable for the complete pipeline.

Component	Suggested technologies	Purpose
Data handling	Python, Pandas, NumPy	Cleaning, validation, window construction and descriptive statistics.
Chart generation	Matplotlib or a financial-chart library	Reproducible K-line image rendering.
Deep learning	PyTorch or TensorFlow	CNN encoder, projection head and supervised classifier.
Clustering	scikit-learn and/or scikit-learn-extra	K-medoids, baseline algorithms and cluster metrics.
Optimisation	Optuna or controlled search scripts	Hyperparameter selection using validation data only.
Experiment tracking	Structured logs, MLflow or Weights & Biases	Record seeds, configurations and performance.
Version control	Git	Track code and configuration changes.
Demonstration	Streamlit, Dash or a lightweight web interface	Display current chart, probabilities and medoids.

Table 13. Suggested implementation technologies.

A recommended software structure is shown below:

```
project/
  data/raw/
  data/processed/
  images/train/ validation/ test/
  src/data_loader.py
  src/window_builder.py
  src/chart_renderer.py
  src/augmentations.py
  src/contrastive_model.py
  src/clustering.py
  src/target_builder.py
  src/predictor.py
  src/evaluation.py
  configs/
  models/
  results/
  reports/
```

11. Risks, Challenges and Mitigation

Risk or challenge	Potential effect	Mitigation
Temporal data leakage	Inflated and unreliable performance	Chronological split before training; training-only clustering; purge boundary-crossing samples.
Clustering by absolute price level	Groups represent historical era rather than structure	Use relative normalisation; remove axis labels; compare normalisation methods.
Invalid image augmentations	Model learns economically altered examples	Prohibit flips, rotations, candle reordering and aggressive cropping.
Cluster imbalance	Accuracy dominated by common structures	Report macro and balanced metrics; use class weighting; examine K.
Cluster instability	Medoids and labels change between runs	Repeat seeds; measure agreement; align clusters by medoid similarity.
Weak transition predictability	Good clusters but poor forecasting	Separate cluster-quality claims from prediction claims; compare with Markov and persistence baselines.
Non-stationarity	New windows do not fit training prototypes	Walk-forward tests; medoid-distance warning; periodic retraining as future work.
Overinterpretation of medoid	Prototype mistaken for exact price path	Display probabilities, alternatives and explicit uncertainty notices.
Overfitting during model selection	Validation-specific performance	Limit search space; lock final design before one-time test evaluation.
Single-asset scope	Limited generalisability	State limitation and recommend multi-asset replication.

Table 14. Principal project risks and planned controls.

12. Ethical, Reproducibility and Practical Considerations

The project uses historical financial-market data and is not expected to involve personally identifiable information. Data licensing conditions and exchange terms must nevertheless be respected.

The model should be presented as a research and decision-support prototype. Cryptocurrency trading involves substantial uncertainty, and historical results do not guarantee future performance. The project should not conceal unsuccessful experiments, unstable clusters or periods of poor performance.

Every reported experiment should state the asset, exchange, interval, study dates, sample construction, split boundaries, number of clusters, random seeds, hyperparameters, transaction-cost assumptions and whether the performance belongs to validation or final test data.

13. Deliverables and Work Plan

13.1 Expected Deliverables

No.	Deliverable	Acceptance evidence
1	Cleaned and documented OHLC dataset	Validation summary, source record and missing-data report.
2	Reproducible 30-candle K-line renderer	Unit tests and sample images.
3	Self-supervised CNN encoder	Saved weights, training curves and embedding diagnostics.
4	Clustering model and medoid catalogue	Cluster metrics, medoid images and descriptive statistics.
5	Next-window supervised classifier	Saved model and calibrated cluster probabilities.
6	Baseline and ablation experiments	Comparative results tables.
7	Chronological out-of-sample evaluation	Classification and structural-similarity report.
8	Demonstration interface	Current frame, predicted clusters, medoids and uncertainty display.
9	Source code and configurations	Version-controlled repository and reproducibility instructions.
10	Final dissertation or technical report	Approved academic submission.

Table 15. Expected project deliverables.

13.2 Indicative 20-Week Plan

Weeks	Main activities
1-2	Finalise asset, interval, source, research questions and literature review.
3-4	Acquire, clean and validate OHLC data; define chronological splits.
5-6	Implement window construction, normalisation and K-line rendering.
7-9	Design financially valid augmentations and train the self-supervised encoder.
10-11	Generate embeddings; select K; evaluate clustering and medoids.
12-13	Construct next-window targets and train baseline classifiers.
14-15	Train and tune the proposed supervised predictor.
16	Run final chronological and structural-similarity evaluation.
17	Conduct ablations and optional paper-trading analysis.
18	Build the demonstration interface and reproduce final results.
19-20	Complete writing, review, corrections and submission preparation.

Table 16. Indicative implementation and reporting schedule.

14. Success Criteria

The project will be considered technically successful when the complete data-to-prediction pipeline is reproducible and when the evaluation provides a clear answer to the research questions. Minimum success does not require guaranteed trading profitability.

- The same configuration and random seed reproduce the reported data splits, chart images and model outputs.
- The embeddings form clusters with measurable coherence and acceptable stability.
- The medoids are observable and meaningfully representative of their assigned members.
- The next-window classifier exceeds the majority baseline and is compared transparently with persistence and Markov baselines.
- All final performance is reported on a strictly later unseen test period.
- The predicted medoids are structurally closer to actual future windows than naive medoid baselines.
- The system reports uncertainty and identifies out-of-distribution windows where possible.
- Limitations and unsuccessful cases are documented.

15. Anticipated Limitations

- K-line images contain the same OHLC information from which they are generated; any advantage arises from representation and architecture.
- The discovered clusters depend on normalisation, rendering, encoder, distance measure and K.
- A medoid is only one representative and cannot express every variation within its cluster.
- The model predicts a broad structure rather than exact future prices.
- Cryptocurrency markets are non-stationary, and new structures may emerge after training.
- A single-asset study cannot establish general performance across cryptocurrencies.
- Strong predictive metrics may not translate into profitable trading after costs and risk are considered.

16. Future Extensions

- Replication across several cryptocurrencies and exchanges.
- Incorporation of volume, order-book, sentiment or macroeconomic information.
- Fusion of visual embeddings with direct numerical OHLC features.
- Use of several preceding windows as a temporal sequence.
- Prediction at multiple future horizons.
- Online or periodically updated clustering and medoid catalogues.
- Explicit out-of-distribution detection.
- Hierarchical prototypes containing broad regimes and finer substructures.
- Comparison with masked-image and non-contrastive self-supervised learning.
- Probabilistic mixtures of several medoids rather than one top prototype.

17. Conclusion

This project proposes a different formulation of cryptocurrency forecasting. Instead of predicting an exact future price or reducing the next period to a simple direction, it seeks to estimate the approximate structural character of the following 30-candle market window.

Historical OHLC values will be converted into standardised K-line images. A contrastive self-supervised CNN will learn embeddings without manually assigned chart-pattern labels. The embeddings will be clustered to discover recurring structural groups, and each group will be represented by its medoid, an actual historical window.

The current window will then be paired with the cluster label of the immediately following non-overlapping window. A supervised classifier will estimate the probable next cluster, and the associated medoid will provide an interpretable visual reference for the predicted dynamics.

The central contribution is the integration of self-supervised chart representation, data-driven market-structure discovery, medoid-based interpretation and supervised next-window transition prediction. The credibility of the work will depend on strict temporal controls, meaningful augmentations, stable clustering, strong baselines and transparent reporting of uncertainty and limitations.

References

- Anscombe, F. J. (1973). Graphs in statistical analysis. *The American Statistician*, 27(1), 17-21. <https://doi.org/10.1080/00031305.1973.10478966>
- Caron, M., Bojanowski, P., Joulin, A., & Douze, M. (2018). Deep clustering for unsupervised learning of visual features. In *Proceedings of the European Conference on Computer Vision* (pp. 132-149). https://doi.org/10.1007/978-3-030-01264-9_8
- Chen, J.-H., & Tsai, Y.-C. (2020). Encoding candlesticks as images for pattern classification using convolutional neural networks. *Financial Innovation*, 6, Article 26. <https://doi.org/10.1186/s40854-020-00187-0>
- Chen, T., Kornblith, S., Norouzi, M., & Hinton, G. (2020). A simple framework for contrastive learning of visual representations. In *Proceedings of the 37th International Conference on Machine Learning* (Vol. 119, pp. 1597-1607). *Proceedings of Machine Learning Research*. <https://proceedings.mlr.press/v119/chen20j.html>
- Hung, C.-C., & Chen, Y.-J. (2021). DPP: Deep predictor for price movement from candlestick charts. *PLOS ONE*, 16(6), e0252404. <https://doi.org/10.1371/journal.pone.0252404>
- Kaufman, L., & Rousseeuw, P. J. (1990). *Finding groups in data: An introduction to cluster analysis*. John Wiley & Sons. <https://doi.org/10.1002/9780470316801>
- Kusuma, R. M. I., Ho, T.-T., Kao, W.-C., Ou, Y.-Y., & Hua, K.-L. (2019). Using deep learning neural networks and candlestick chart representation to predict stock market. *arXiv*. <https://doi.org/10.48550/arXiv.1903.12258>
- Oestreicher, C. (2007). A history of chaos theory. *Dialogues in Clinical Neuroscience*, 9(3), 279-289. <https://doi.org/10.31887/DCNS.2007.9.3/coestreicher>
- Rodrigues, F., & Machado, M. (2025). High-frequency cryptocurrency price forecasting using machine learning models: A comparative study. *Information*, 16(4), Article 300. <https://doi.org/10.3390/info16040300>
- Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53-65. [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7)
- Zhu, Y., Ma, J., Gu, F., Wang, J., Li, Z., Zhang, Y., Xu, J., Li, Y., Wang, Y., & Yang, X. (2023). Price prediction of Bitcoin based on adaptive feature selection and model optimization. *Mathematics*, 11(6), Article 1335. <https://doi.org/10.3390/math11061335>